Real-time Simulation and CHiL Testing of Power System Stabilizers on Microcontrollers with Modelica and eFMI

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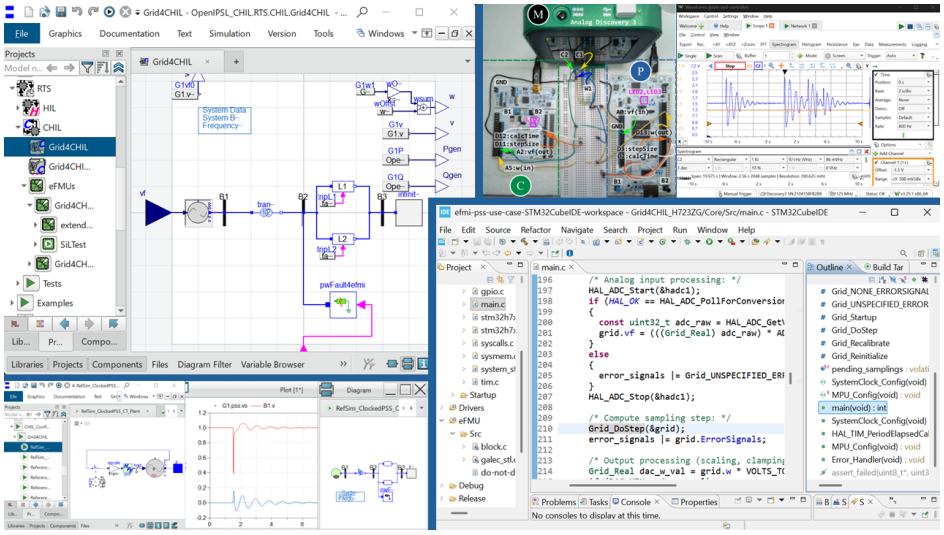


Figure 1. *Top left & bottom right: Modelica model & derived eFMI STM32 firmware in Dymola & STM32CubeIDE; Bottom left & top right: MiL simulation of the CHiL setup & actual measurements in Digilent WaveForms.*

Challenge. Modern power grids face significant stability challenges, as shown by recent incidents such as the 2025 oscillations of the Iberian grid, highlighting the need for robust model-based systems engineering (MBSE) to enable rapid control redesign and deployment. Yet, the move to controller hardware-in-the-loop (CHiL) testing is hindered by manual re-implementation, costly and complex tooling stacks, and the difficulty of obtaining real-time suited plant models—all of which disrupt traceability.

Presented solution. This paper pioneers a workflow using the Modelica® language and the eFMI® (Functional Mock-up Interface for embedded systems) Standard to bridge this gap in current practice within the power grid domain (see Figure 1). We introduce the `OpenIPSL_CHIL` Modelica library, extending the Open-Instance Power System Library (`OpenIPSL`) for embedded real-time applications: A clocked power system stabilizer (PSS) controller *and* the generator-with-grid plant become real-time-suited models, keeping physics as the single source of truth. The eFMI tooling of Dymola® in turn synthesizes both into production-ready C17 code (fixed 0.2 ms/5 kHz sampling, inline explicit-Euler integration), system-integrated using STM32Cube tooling for IO and clock-tree/timer configuration and automatic hardware abstraction layer (HAL) code generation, and deployed on two low-cost STM32® development boards with ARM® Cortex®-M microcontrollers wired in closed loop: A NUCLEO-L476RG (80 MHz) for the controller and a NUCLEO-H723ZG (550 MHz) for the plant.

Validation results. The workflow is validated by a full suite of model-, software- and controller hardware-in-the-loop (MiL, SiL, CHiL) tests. In CHiL time-domain tests the engaged PSS damps the generator speed and recovers from ± 0.2 pu deviations, whereas the uncompensated plant becomes unstable beyond ± 0.1 pu; measured open- and closed-loop frequency responses (0.15 Hz to 100 Hz) match the Dymola-linearized models. Non-functional quality is verified too: MISRA C:2023 and SEI CERT C Coding Standard compliance, strict eFMI GALEC error signaling excluding numerical issues, and measured execution times confirming real-time capability with ample headroom.

Take-aways. The workflow is a traceable and cost-effective alternative to proprietary real-time platforms, replacing setups typically costing $\gg$\$50,000 with $<$\$50 microcontrollers. All source models and generated artifacts are released under the 3-Clause BSD License at `https://github.com/ALSETLab/efmi-pss-use-case`, providing an open-source MBSE workbench for equation-based design, MiL-CHiL validation, and microcontroller deployment of power system stabilizers (PSS).

Keywords: power systems, power system stabilizer, real-time simulation, CHiL, embedded control, Modelica, eFMI, Dymola, STM32, ARM Cortex-M